Computer Vision Homework 1

NTU CSIE D13922014 黃丰楷

Part 1

(a) Upside-down lena.bmp

- Use a for-loop to reorder the value of all rows.
- Code

```
ud_img = np.empty(img.shape)
for i in range(img.shape[0]):
    ud_img[img.shape[0]-i-1,:] = img[i,:]
```

- Result

(b) Right-side-left lena.bmp

- Use a for-loop to reorder the value of all columns.

- Code

```
rl_img = np.empty(img.shape)
for i in range(img.shape[1]):
    rl_img[:,img.shape[1]-i-1] = img[:,i]
```

- Result

(c) Diagonally flip lena.bmp

- Use two for-loops to reorder the value of all rows and columns.
- Code

```
diag_img = np.empty(img.shape)
for i in range(img.shape[0]):
    for j in range(img.shape[1]):
        diag_img[img.shape[0]-i-1,img.shape[1]-j-1] = img[i,j
```

- Result

Part 2

(a) Rotate lena.bmp 45 degrees clockwise

- Find the rotation matrix centered with the center of the original image and project the original image with the rotation matrix to obtained the rotated one.
- Code

```
(h, w) = img.shape[:2]

center = (w // 2, h // 2)
M = cv2.getRotationMatrix2D(center, -45, 1.0)

cos = np.abs(M[0, 0])
sin = np.abs(M[0, 1])

new_w = int((h * sin) + (w * cos))
new_h = int((h * cos) + (w * sin))

M[0, 2] += (new_w / 2) - center[0]
M[1, 2] += (new_h / 2) - center[1]
```

```
rot_img = cv2.warpAffine(img, M, (new_w, new_h))
cv2.imwrite('./output/rotated_img.jpg',rot_img)
```

- Result

(b) Shrink lena.bmp in half

- Resize the original image in half and paste it to the upper left corner of a white image that has the same size as the original image.
- Code

```
resize_img = cv2.resize(img,(img.shape[0]//2,img.shape[1]//2
shrinked_img = np.zeros(img.shape)
shrinked_img[:resize_img.shape[0],:resize_img.shape[1]] = res
cv2.imwrite('./output/shrinked_img.jpg',shrinked_img)
```

- Result

(c) Binarize lena.bmp at 128 to get a binary image

- Assign the pixel in the original image that greater than 128 to be 255, else to be 0.
- Code

```
binary_img = img.copy()
binary_img[np.where(img>128)] = 255
binary_img[np.where(img<=128)] = 0
cv2.imwrite('./output/binarized_img.jpg',binary_img)
```

- Result